

Lab# 01: Analyse OS & Application Logs

Objective: Learn to analyse OS and Application Logs

Understanding system and application logs is crucial for troubleshooting system issues, identifying security threats, and monitoring system health. Most logs have levels (e.g., Info, Warning, Error). Focus on higher levels for troubleshooting and security monitoring. Use tools' built-in filtering or search functionalities to narrow down log entries. Look for patterns or events occurring around the same time that might be related. If you find an error message you don't understand, search online for explanations and solutions.

a.) Windows OS Logs

Example 1: Identifying Application Crash

- i. Open **Event Viewer** (build-in tool), search for it in the Start Menu.
- ii. Expand **Windows Logs** and select **Application**.
- iii. Look for events with a Critical or Error level. These might indicate application crashes.
- iv. Double-click the suspicious event to see details. The description will often point to the source of the issue.

Example 2: Monitoring Security Events

- i. Expand **Windows Logs** and select **Security**.
- ii. Look for events with Failure status. These might indicate unauthorized access attempts.
- iii. Double-click the event to see details like source, user involved, and the specific action attempted.

b.) Linux OS Logs

System processes, applications, and security tools running on Kali create log messages. These messages are sent to '**rsyslog**', which activates based on its configuration. rsyslog can filter out irrelevant messages and apply formatting to make logs easier to understand. Logs can be directed to local files on the Kali system itself or forwarded to a remote logging server for centralized management. Rsyslog (Rocket-Fast System for Log processing) is a powerful and versatile message logging utility for Linux distributions, including Kali Linux, offering significant advantages over the traditional 'syslogd' daemon.

- **Enhanced Performance:** rsyslog boasts superior speed and efficiency in handling system logs, making it ideal for resource-constrained environments like penetration testing setups
- **Modular Design architecture** allows for customization and extension through modules, enabling advanced logging capabilities like message filtering, output formatting, and remote forwarding.
- **Security Focus:** rsyslog incorporates security features like support for secure protocols (TLS/SSL) when transmitting logs to remote servers, safeguarding sensitive information.

While **systemd-journald** is now the default logging service in newer Kali versions (based on Debian 12 and up), rsyslog is better option for managing and analyzing system logs in Kali Linux, especially in penetration testing scenarios where centralized logging, advanced filtering, and secure remote log forwarding are crucial.

Steps:

- i. Linux logs can be found in /var/log folder. In case logs are not visible, install 'rsyslog'.

```
(kali㉿kali)-[/var/log]
└─$ sudo apt-get install rsyslog
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libestr0 libfastjson4 liblognorm5
Suggested packages:
  rsyslog-mysql | rsyslog-pgsql rsyslog-mongodb rsyslog-doc rsyslogd
The following NEW packages will be installed:
  libestr0 libfastjson4 liblognorm5 rsyslog
0 upgraded, 4 newly installed, 0 to remove and 1179 not upgraded.
```

- ii. Open a terminal window go to /var/log folder and view the files and folders.

```
(kali㉿kali)-[~]
└─$ cd /var/log

(kali㉿kali)-[/var/log]
└─$ ls -l
total 1924
-rw-r--r-- 1 root root 597 Jun 14 11:54 alternatives.log
-rw-r--r-- 1 root root 8026 May 11 20:16 alternatives.log.1
-rw-r--r-- 1 root root 177 Apr 8 15:19 alternatives.log.2.gz
-rw-r--r-- 1 root root 3717 Mar 17 09:16 alternatives.log.3.gz
-rw-r--r-- 1 root root 1768 Dec 4 2023 alternatives.log.4.gz
-rw-r--r-- 1 root root 9753 Nov 17 2023 alternatives.log.5.gz
drwxr-x--- 2 root adm 4096 Mar 18 08:44 apache2
drwxr-xr-x 2 root root 4096 Jun 14 12:05 apt
-rw-r--r-- 1 root root 19857 Jun 15 11:56 boot.log
-rw-r--r-- 1 root root 34758 Jun 14 10:59 boot.log.1
```

Use Case 1: Identify Login Attempts

Step 1: Use the command `grep Client vsftpd.log.1` in the terminal window. This searches the vsftpd application log for lines containing "Client" (indicating access and login attempts).

```
(kali㉿kali)-[/var/log]
└─$ sudo grep client vsftpd.log.1

(kali㉿kali)-[/var/log]
└─$ sudo grep Client vsftpd.log.1
Wed Mar 20 03:35:39 2024 [pid 41955] CONNECT: Client "::1"
Wed Mar 20 03:35:43 2024 [pid 41954] [kali] OK LOGIN: Client "::1"
Wed Mar 20 03:35:56 2024 [pid 42394] CONNECT: Client "::1"
Wed Mar 20 03:36:02 2024 [pid 42393] [kali] OK LOGIN: Client "::1"
Wed Mar 20 03:39:50 2024 [pid 47806] CONNECT: Client "::ffff:192.168.119.138"
Wed Mar 20 03:39:54 2024 [pid 47805] [kali] OK LOGIN: Client "::ffff:192.168.119.138"
Wed Mar 20 03:42:38 2024 [pid 51537] CONNECT: Client "::1"
Wed Mar 20 03:42:41 2024 [pid 51536] [kali] OK LOGIN: Client "::1"
Wed Mar 20 03:46:50 2024 [pid 57457] CONNECT: Client "::ffff:127.0.0.1"
Wed Mar 20 03:46:50 2024 [pid 57456] [kali] OK LOGIN: Client "::ffff:127.0.0.1"
```

Step 2: Analyze the output to view timestamps, usernames, and if login attempt was successful or failed.

Step 3: We can also check for login or failure in 'auth.log' file

```
(kali㉿kali)-[/var/log]
└─$ sudo grep login auth.log
2024-06-15T12:25:33.258830+06:00 kali systemd-logind[553]: New session c2 of user lightdm.
2024-06-15T13:29:57.749231+06:00 kali systemd-logind[553]: Removed session c2.
2024-06-15T13:31:20.163397+06:00 kali sudo:    kali : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/grep login auth.log

(kali㉿kali)-[/var/log]
└─$ sudo grep fail* auth.log
2024-06-15T13:31:33.747029+06:00 kali sudo:    kali : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/grep faillog auth.log
```

```
(kali㉿kali)-[/var/log]
└─$ sudo grep 'password' vsftpd.log.1
Sat Mar 29 06:40:51 2025 [pid 40279] [ftp] OK LOGIN: Client "::ffff:192.168.119.154", anon password "?"
Sat Mar 29 06:49:57 2025 [pid 44681] [ftp] OK LOGIN: Client "::ffff:192.168.119.154", anon password "?"
```

Step 4: Check for Login attempts into a locally hosted web site as Kali has a preinstalled Web Server (Apache2 Web server)

```
(kali@kali)-[/var/log/apache2]
$ sudo grep 'password' access.log
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /change_password HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /change_password HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /forgot_password HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /forgot_password HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /forgot_password HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /forgot_password HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /lostpassword HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"

(kali@kali)-[/var/log/apache2]
$ sudo grep 'client' access.log
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /aspnet_client HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /client HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /clientaccesspolicy HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /clientapi HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /clientes HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
::1 - - [24/Jun/2025:23:10:48 -0400] "GET /clients HTTP/1.1" 404 432 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1)"
```

Use Case 2: Check User Login Management

Step 1: Use **'sudo grep login boot.log'** in the terminal window to determine user logins

```
(kali@kali)-[/var/log]
$ sudo grep kali boot.log

(kali@kali)-[/var/log]
$ sudo grep login boot.log
[ OK ] Started systemd-logind.service - User Login Management.
Starting systemd-logind.service - User Login Management ...
[ OK ] Started systemd-logind.service - User Login Management.
Starting systemd-logind.service - User Login Management ...
[ OK ] Started systemd-logind.service - User Login Management.
```

Use Case 3: Monitor System Activity (using 'tail' and '/var/log/syslog')

Step 1: Use the command **'tail -f /var/log/syslog'** to display the last lines of the system log file.

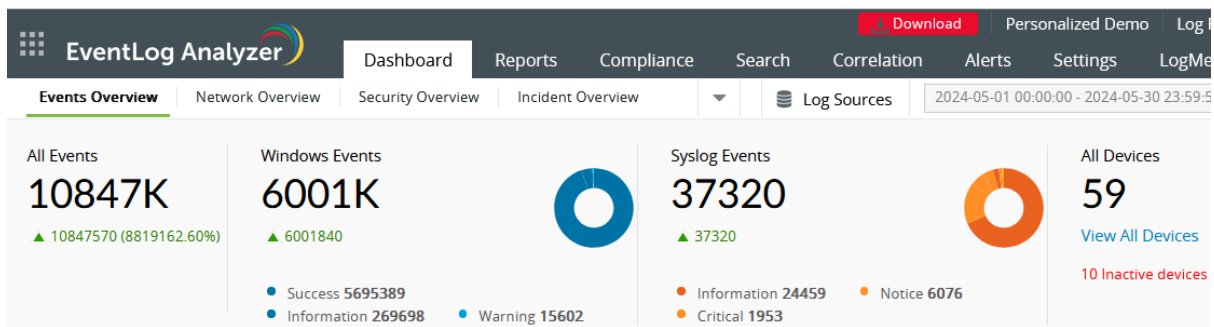
Step 2: Watch the output for suspicious activity or error messages and press 'Ctrl+C' to stop when done.

```
(kali@kali)-[/var/log]
$ sudo tail -f /var/log/syslog
2024-06-15T12:11:11.608992+06:00 kali dbus-daemon[539]: [system] Activating via systemd: s
patcher.service' requested by ':1.5' (uid=0 pid=641 comm="/usr/sbin/NetworkManager --no-da
2024-06-15T12:11:11.633782+06:00 kali systemd[1]: Starting NetworkManager-dispatcher.servi
2024-06-15T12:11:11.662530+06:00 kali dbus-daemon[539]: [system] Successfully activated se
2024-06-15T12:11:11.662744+06:00 kali systemd[1]: Started NetworkManager-dispatcher.servic
2024-06-15T12:11:21.688250+06:00 kali systemd[1]: apt-daily-upgrade.service - Daily apt up
onditionACPower=true).
2024-06-15T12:11:21.693443+06:00 kali systemd[1]: NetworkManager-dispatcher.service: Deact
2024-06-15T12:15:01.595560+06:00 kali CRON[42125]: (root) CMD (command -v debian-sa1 > /de
2024-06-15T12:16:53.114898+06:00 kali dbus-daemon[3356]: [session uid=1000 pid=3356] Activ
comm="xfce4-panel --display :0.0 --sm-client-id 242fc32f")
2024-06-15T12:16:53.131485+06:00 kali dbus-daemon[3356]: [session uid=1000 pid=3356] Succe
2024-06-15T12:17:01.611322+06:00 kali CRON[46266]: (root) CMD (cd / && run-parts --report
```

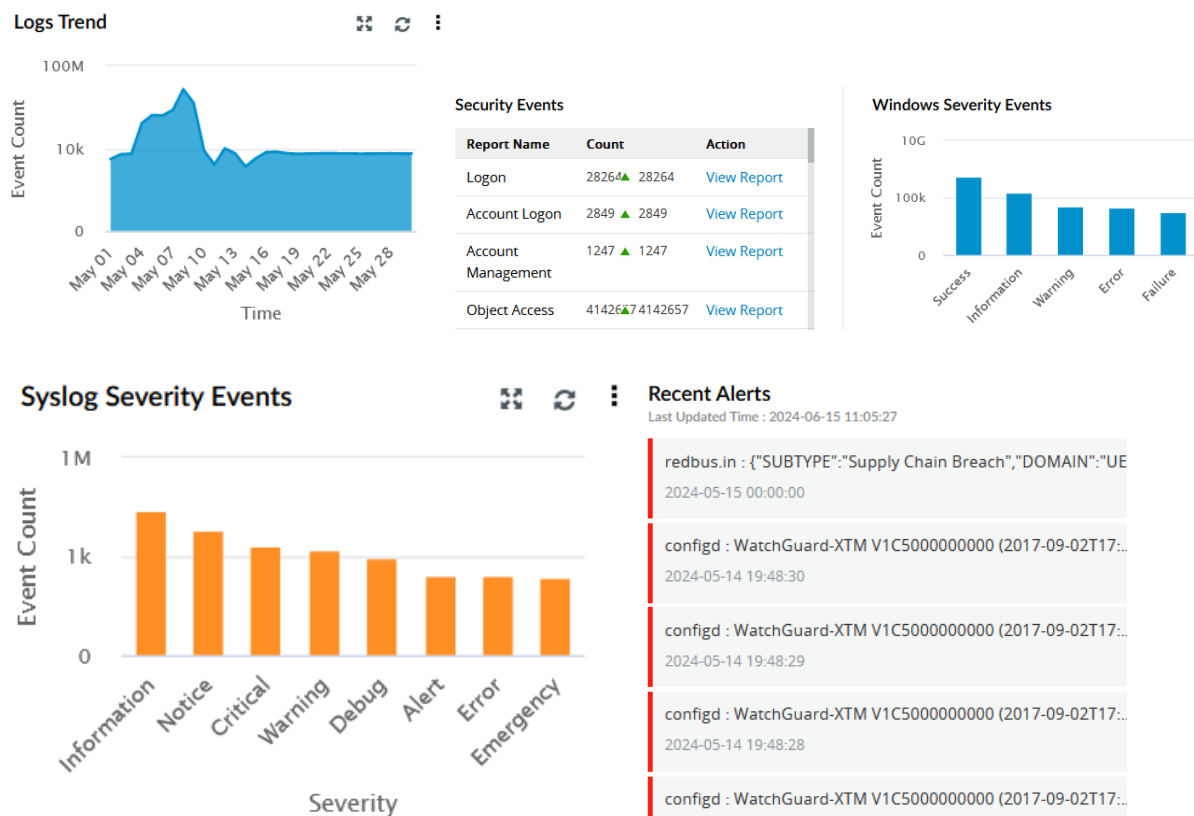
c.) Demo Event Log Dashboard

We have looked at RAW logs using OS utilities. Imagine 10,000 systems and their logs in an IT setup. These are accessed using a centralized dashboard by System Admins to search for events/incidents.

Step 1: Access the demo dashboard for Manage Engine Event Log Analyzer from <https://demo.eventloganalyzer.com/event/index2.do?url=emberapp#/home/dashboard/301>



Step 2: This displays the commercial dashboard for Events, Trends, Severity Alerts for app/device logs



d.) Sample Web Log analysis

System Admins also perform Web Log analysis to gather information and intelligence.

Step 1: Study the sample web application log from <https://www.weblogexpert.com/sample/index.htm>

Contents	
+  General Statistics	
+  Activity Statistics	
+  Access Statistics	
+  Visitors	
+  Referrers	
+  Browsers	
+  Errors	
+  Tracked Files	

Hits	
Total Hits	3,135
Visitor Hits	3,059
Spider Hits	76
Average Hits per Day	447
Average Hits per Visitor	6.26
Cached Requests	0
Failed Requests	28
Page Views	
Total Page Views	957
Average Page Views per Day	136
Average Page Views per Visitor	1.96
Visitors	
Total Visitors	489
Average Visitors per Day	69

Step 2: Commercial log dashboard present log activities for Access, Visitors, Browsers and Errors to understand the application log trends and present them to the management.

e.) Install Free Log Analyser

Graylog is a centralized logging solution that allows the user to aggregate and search through logs. It provides a powerful query language, a processing pipeline for data transformation, and alerting abilities.

Step 1: Download Graylog docker image in Linux from <https://hub.docker.com/r/graylog/graylog>

Step 2: Refer to the Graylog Docker documentation steps from https://go2docs.graylog.org/current/downloading_and_installing_graylog/docker_installation.htm

Step 3: Windows Event IDs: Refer to <https://graylog.org/post/critical-windows-event-ids-to-monitor/> to study about Windows Event IDs.

f.) Web Server - Apache

Step 1: Copy log samples for normal server functions from https://www.ossec.net/docs/log_samples/apache/apache.html#

Step 2: Save the log samples to a CSV file and upload them to Graylog and analyze.

Step 3: Copy log samples for web attacks from https://www.ossec.net/docs/log_samples/apache/apache_attacks.html

Step 4: Save the attack log samples and upload them to Graylog and analyze.

g.) SQL Database Logs

Step 1: Copy log samples to a CSV file from https://www.ossec.net/docs/log_samples/apache/apache_attacks.html#sql-injection-attempt-on-php-nuke

Step 2. Upload the CSV file and upload them to Graylog and analyze.

h.) Manage Engine Event Log Demo

Step 1: Download Trial version from <https://www.manageengine.com/products/eventlog/request->

[demo.html](#)

i.) SolarWinds Loggly

Step 1: Register a 30-day Cloud-based account from <https://www.loggly.com/signup/>

Lab activities:

- **Perform the above-mentioned steps and present your work in WORD file with screenshots snipped with relevant details.**
- **Do NOT copy from others, else your submission will be graded ZERO.**